PRATIKSHA PENDSE

SOFTWARE ENGINEER – EMBEDDED SYSTEMS, C & HARDWARE

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ResearchGate



Embedded software engineer with an M.Sc. in Automotive Software Engineering and a background in electronics. At Rohde & Schwarz, I developed a real-time oscilloscope detector from a MATLAB model to a VHDL design verified in simulation, working across signal processing, architecture and testing. My embedded projects include a C I^2C sensor driver on PREEMPT_RT Linux and C/C++ robots tested on hardware. I would like to bring this model-to-hardware approach to embedded control development.

SKILLS

Programming	C, C++, Python, MATLAB, VHDL
Embedded	Arduino, ESP32, Raspberry Pi, STM32 (basics), sensor interfacing, FPGA, PREEMPT_RT Linux
Methods	Model-based development, V-model, test-driven development, CI/CD
Tools	Git, GitLab, Linux, Questa, ROS, Gazebo
Languages	English C1, German B2 (in progress)

EDUCATION

M.Sc. Automotive Software Engineering, Chemnitz University of Technology — Grade 2.0 — Oct 2022 – Sept 2026
Modules: Embedded Software Design, HW/SW Codesign, Model-Driven Development, Automotive Sensor Systems

B.E. Electronics and Communication Engineering, KLS Gogte Institute of Technology, India — CGPA 9.11 — Aug 2017 – Aug 2021
Modules: Advanced Processors + lab, Control Systems, DSP & Architecture, GPS & IoT, Sensors, C/C++ labs

EXPERIENCE

Master Thesis Student, Signal Processing – Oscilloscopes — Sept 2025 – July 2026
Rohde & Schwarz GmbH & Co. KG, Chemnitz & TU Chemnitz, Chair of Computer Engineering

- Converted a user problem, unreliable decimated waveforms, into a real-time aliasing detector for oscilloscopes.
- Surveyed the state of the art and designed the hardware/software system architecture of the solution.
- Modelled the detector in MATLAB, implemented it in VHDL and verified the two against each other in simulation.
- Analysed accuracy-versus-resource trade-offs and identified possible limitations and directions for future work.

Working Student, Software Tool Development and Testing — Feb 2025 – Aug 2025
Rohde & Schwarz GmbH & Co. KG, Chemnitz

- Built a web dashboard that compares test results across devices and software versions for the development teams.
- Replaced manual report comparison with interactive online plots, so developers see differences at a glance.
- Set up real-time temperature tracking for the test lab, with automatic Microsoft Teams alerts via webhooks.

Intern, Software Tool Development — June 2024 – Nov 2024
Rohde & Schwarz, Dresden

- Co-developed an object-oriented Python framework validating ASIC/FPGA system descriptions for code generation.
- Designed layered validation and cross-component checks with Pydantic models and NetworkX dependency graphs.
- Output feeds MATLAB, SystemC and VHDL code generators; delivered test-first through GitLab CI/CD.

Student Assistant, Computer Vision and Robotics — Apr 2023 – Oct 2024
Technische Universität Chemnitz

- Implemented LiDAR-based human tracking in ROS and C++, validated in Gazebo simulation.

Systems Engineer, ITSM Development — Aug 2021 – Jan 2023
Tata Consultancy Services, India

- Developed ITSM workflows, integrations and custom web front-ends, and trained internal teams on adoption.

PROJECTS

- **Smart ambulance & patient monitoring (ESP32, B.E. project)** – traffic management; patient data sent to hospitals via ThingSpeak.
- **Real-time sensor driver (Raspberry Pi)** – I2C IMU driver written in C as a kernel module on PREEMPT_RT Linux.
- **Robots (Arduino)** – line-follower, voice-controlled and object-following robots, programmed in C/C++ and tested on hardware.
- **Home automation (Arduino, Raspberry Pi)** – remote control of home appliances.